

Research Method

This research will follow a qualitative and experimental research methodology consisting of literature analysis, model development, and evaluation.

Step 1 – Literature Review

Research papers related to Sinhala OCR[3, 4] , Handwriting recognition, GAN-based font generation, and deep learning approaches for non-Latin scripts[5] will be reviewed to identify existing methods, limitations, and research gaps.

Step 2 – Sinhala Script Analysis

The structural properties of Sinhala characters, including compound glyphs, vowel modifiers, and stroke patterns[6], will be analyzed to understand challenges in font generation[7].

Step 3 – Dataset Collection and Preparation

Handwritten Sinhala character samples will be collected from multiple users after obtaining their informed consent. The collected samples will then be preprocessed using image normalization, resizing, and noise removal techniques.

Step 4 – Model Selection and Development

Suitable deep learning architectures such as:

- Generative Adversarial Networks (GANs)
- Conditional GANs (CGANs) [8]
- Component-Guided GANs (CG-GANs) [9]

will be studied and adapted for Sinhala font generation.

Step 5 – Font Generation

The developed model will generate Sinhala glyphs based on limited handwriting samples while preserving writing style consistency.[9]

Step 6 – Evaluation

Generated fonts will be evaluated using:

- Readability
- Structural consistency
- Visual similarity to original handwriting
- User feedback

